



Board Governance and Sustainability Disclosure Quality in Integrated Reports: Evidence From the Johannesburg Stock Exchange Integrated Reporting Awards

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Abstract: This study examines the association between board structure and characteristics and the quality of sustainability reporting, using an integrated reporting-based proxy. Thirty-seven companies listed on the Johannesburg Stock Exchange (JSE) in South Africa were selected. Panel data were collected on board size, women on the board, ethnic diversity, the number of financial experts on the board, the average age of board members and their independence. Three control variables were included, namely profitability, firm age and firm size. Sustainability reporting quality (SRQ) was operationalised using the Ernst and Young (EY) Excellence in Integrated Reporting Awards, a categorical rating with four levels: Progress to be made, Average, Good, and Excellent. A multinomial logit model with firm level clustered standard errors was applied to assess the relationship between board structure and SRQ. Under the sample conditions, the results indicate a statistically significant positive association between ethnic diversity and the highest SRQ category. Board size and average board age showed negative and positive associations respectively, but neither was statistically significant after clustering. Board independence, financial expertise and gender diversity were not statistically associated with SRQ. The findings offer insights for policymakers on board composition, with ethnic diversity emerging as the characteristic most strongly associated with high-quality sustainability disclosure under the sample conditions examined.

Keywords: Board governance characteristics; Board structure; Emerging markets; Integrated reporting; Multinomial logistic regression; South Africa; Sustainability reporting quality

1. Introduction

The concept of sustainability reporting has evolved and grown in prominence since the introduction of the Brundtland report (Brundtland, 1987). Several sustainability reporting initiatives have since been implemented. According to Sabelfeld et al. (2024), various examples of such initiatives include, amongst others, the Global Reporting Initiative's (GRI) Standards, the Integrated Reporting Framework, and an arguably less formal framework, the United Nations Sustainable Development Goals (UN SDGs). These frameworks provide guidance for reporting on environmental, social and economic impacts and on how organisations' strategy, governance and performance have led to value creation over time (Bose, 2020; Luque-Vílchez et al., 2023; Siew, 2015).

Sustainability reporting is often explained through the lens of the stakeholder theory, which has evolved from the work of various scholars over time (Freeman, 2010; Joseph, 2012; Miles, 2017). Mahajan et al. (2023) explained that stakeholder theory involves acknowledging all stakeholders, both internal and external, in an organisation's activities. Stakeholders have embraced this and moved to exert pressure on organisations to report on more than economic performance, and to include information on their environmental and social practices (Benameur et al., 2024; Siew, 2015).

A growing trend towards sustainable investment has engendered the need for sustainability reports amongst

investors, as they seek to make more informed, responsible investment decisions. Although sustainability reports are not mandatory in most countries, many firms have widely adopted the issuance of these disclosures as an integral part of their corporate reports (Benameur et al., 2024; Dilling, 2010). For instance, many EU member states have implemented mandatory reporting under the recently introduced Corporate Sustainability Reporting Directive (CSRD), a regulatory framework designed to enhance corporate transparency and accountability in sustainability reports (Hristov & Searcy, 2024).

For most African countries, the adoption of sustainability disclosures is still limited and remains largely voluntary (ACCA, 2014; Wachira et al., 2020). In South Africa, environmental, social, and governance (ESG) disclosure requirements are stricter on public and state entities and on Johannesburg Stock Exchange (JSE) listed companies, which are obligated to disclose financially material ESG issues as part of the JSE listing requirements. For other entity types, ESG disclosure requirements are largely voluntary (Davids et al., 2025).

There has been a heightened interest in various aspects of research focusing on sustainability reporting and corporate disclosures (Al-Qudah & Houcine, 2024; Klettner et al., 2014). Prior research has largely concentrated on assessing the quality of institutional frameworks and the influence of stakeholder pressures on organisations (AbuRaya, 2024). Concurrently, other scholars have attempted to examine the link between various firm-specific characteristics and sustainability reporting (Balasubramanian et al., 2021; Bhatia & Tuli, 2017; Dilling, 2010). Of these studies, many have been skewed towards understanding the impact of financial and quantitative firm-specific characteristics on sustainability issues.

Despite the growing body of sustainability reporting literature, existing findings on the relationship between board governance characteristics and sustainability reporting quality (SRQ) remain fragmented and inconclusive (Al-Qudah & Houcine, 2024). Prior studies have produced mixed evidence regarding the influence of diversity, board structure, and governance characteristics on sustainability disclosures (Amran et al., 2014; Correa-Garcia et al., 2020), while much of the existing evidence remains concentrated in developed markets. Furthermore, limited attention has been given to emerging market contexts such as South Africa, where sustainability reporting practices operate within a distinctive governance and institutional environment shaped by governance reforms, stakeholder-oriented reporting practices, and transformation policies (Bezuidenhout et al., 2007; Institute of Directors South Africa, 2016).

To fill this gap, the objective of the study is to examine the association between various board characteristics and SRQ in the Top 40 companies of the JSE. Understanding the relationship of board characteristics and corporate governance attributes on SRQ is expected to contribute to the limited body of literature exploring the relationships between qualitative variables and SRQ.

The study focuses on South Africa companies to contribute to an emerging market perspective on the different firm variables that influence SRQ. Furthermore, South Africa's ranking as a global leader in corporate governance-related matters, including in integrated reporting, makes it an interesting and relevant case for understanding the impacts of its governance reforms on SRQ (Banhegyi, 2007; Kana, 2020).

South Africa has undergone legislative reforms to address challenges of the lack of inclusivity on boards following its unique apartheid history (Nel et al., 2022). This presents the relevant setting to address the question of whether corporate governance reforms regarding board composition and diversity influence the quality of sustainability disclosures on JSE-listed companies.

The composition of the board is perceived to be an important characteristic, considering the critical role executed by the board of directors in corporate governance. Since the release of the Tyson Report (Tyson, 2003), many corporate governance frameworks have placed great importance on the role of board diversity in enhancing the effectiveness of the board of directors. The South African King IV Code of Governance advocates for boards to be diverse in terms of gender, race, skills, experience and perspectives (Institute of Directors South Africa, 2016).

Some scholars suggest that increased diversity enhances the roles of scrutiny and monitoring performed by the board of directors (Mans-Kemp & Van der Lugt, 2020). A few studies have identified the agency theory as a major influence on an organisation's sustainability reporting. These studies posit that the monitoring role of the board motivates firms to disclose information in attempts to minimise agency costs and information asymmetry (Belal & Roberts, 2010; Moses et al., 2020; Shamil et al., 2014; Vitolla et al., 2020).

Other studies have employed the legitimacy theory to explain the link between corporate governance and sustainability reporting. These studies suggest that firms will seek to function within the ambit of societal norms, values and expectations. Consequently, they will gravitate towards aligning their actions and disclosures to conform to what stakeholders perceive as acceptable (Ali et al., 2021; Hahn & Kühnen, 2013; Herbert & Graham, 2022). The board of directors is tasked with deciding on an organisation's strategic direction. Consequently, the boards of directors are therefore positioned to prioritise sustainability practices that both align the firm with societal norms and expectations and communicate that alignment to stakeholders through sustainability reporting.

The contribution of this study is to add to the body of knowledge on the influence of firm-specific governance characteristics on sustainability reporting in the context of emerging markets. Furthermore, the study employs both the agency and legitimacy theories to explain any links between board characteristics and sustainability reporting. The use of a multi-theory contribution is considered important, given suggestions that the multifaceted nature of

sustainability reporting compels the use of a multi-theoretic approach instead of a single theory approach, which is considered insufficient (Cormier et al., 2005).

2. Literature Review and Hypotheses Development

Sustainability reporting communicates how an entity's ESG principles impact its activities (GRI, 2011). The sustainability report assists various stakeholders in obtaining clarity on the effects of a company's operations on the environment and wider society. Furthermore, it serves as a tool that helps stakeholders assess the risks and opportunities companies face and judge the sincerity of an entity's actions.

The South African ESG landscape is unique from many countries. The country's progression in sustainability reporting is partly due to its political history and transition from apartheid to a multi-racial democracy (Bezuidenhout et al., 2007; Formanek, 2021; GRI, 2013). Increasing pressure from investors to withdraw their investments during apartheid forced local and international corporations to respond with voluntary initiatives, such as the Sullivan principles, to address social injustices (Bezuidenhout et al., 2007; GRI, 2013). Later legislation was built in through Black Economic Empowerment and now the Broad-Based Black Economic Empowerment (B-BBEE) (Clayton et al., 2015).

In substance, not all ESG reporting requirements are voluntary within the South African landscape (Ngorima, 2019). ESG principles are built in as requirements for several business actions. Failure to adopt specific ESG principles may impede a company's ability to access capital markets and lucrative business opportunities (MacNeil & Esser, 2022). Public and state entities, together with companies seeking to obtain a listing on the JSE, are required to adhere to certain ESG principles (Davids et al., 2025; Greenstone et al., 2023). Furthermore, to be eligible to conduct business with public and state institutions, private enterprises are required to conform to social and governance norms, such as the B-BBEE legislation (Ball, 2021).

While sustainability reporting is often associated with voluntary corporate accountability and stakeholder responsiveness, the South African context reflects a more complex interaction between voluntary disclosure practices and institutional governance pressures. Governance reforms such as B-BBEE legislation, King IV, and JSE disclosure requirements have embedded inclusivity and ESG-related considerations into corporate governance structures. Consequently, SRQ within South Africa may reflect not only normative organisational commitment, but also legitimacy-seeking behaviour and compliance with evolving regulatory and governance expectations. This aligns with legitimacy theory, which suggests that firms align disclosure practices with societal and institutional expectations to maintain organisational legitimacy.

The King IV Code of Governance is a radical guidance framework that incorporates ESG requirements in the activities of various companies. Companies listed on the JSE are obligated to comply with the corporate governance requirements of the King IV Code on an 'apply or explain' basis. It challenges directors to integrate thinking through considering the connectivity and interdependence between several factors that affect a firm's ability to create value over time. It emphasises awareness that companies are integral to society and that society is affected by their actions. Against this background, the King IV code recommends that companies publish integrated reports which reflect on the entity's financial and non-financial performance, including sustainability performance (Institute of Directors South Africa, 2016).

The integrated report is a form of sustainability reporting based on the guidelines of the International Integrated Reporting Council (IIRC). The IIRC was established in 2010 by a coalition of organisations, including the GRI, HRH the Prince of Wales' Accounting for Sustainability Project (A4S), and the International Federation of Accountants (IFAC) (Shoaf et al., 2018). Integrated reporting combines financial and non-financial performance into a single report to reflect holistically how financial and ESG criteria contribute to value creation (Dimes & Molinari, 2024). Many companies on the JSE issue a separate sustainability report; however, following the recommendations of the King IV code, most report their ESG activities in the integrated report. In this study, all companies in the sample publish an integrated report, naturally covering various aspects of ESG performance.

2.1 Sustainability Reporting Quality

There has been an evident increase in sustainability reporting amongst companies; however, this has not necessarily translated to quality disclosures (Junior et al., 2014). There is a concern that managers are more likely to disclose information that will enhance the company's image rather than that which will provide a transparent and accountable response to stakeholders (Owen et al., 2000). There is a wide variety of information covered in sustainability reports, including topics on employees, customers, and the environment. Some reports focus on the quantity of information disseminated as opposed to its usefulness to stakeholders (Chang et al., 2017; Junior et al., 2014). There appears to be a consensus in the literature that the quality of a sustainability report should enable stakeholders to fully comprehend the impact that the company's value creation process has on the environment and on society. Eccles & Krzus (2014) argued that the significance of integrated reporting lies less in how many firms adopt the practice than in how well they adopt it, since the usefulness of such reports to investors depends

on how competently the underlying frameworks and standards are applied. Different frameworks guide sustainability reporting practices; however, quality reporting depends less on mechanical adherence to a framework than on the mindset brought to it, requiring a longer-term and more operational outlook that reshapes how firms think about and monitor performance (Pistoni et al., 2018). It could be argued that a high-quality sustainability report would be considered more substantive than symbolic.

The companies in our sample all publish integrated reports and hence report on various ESG areas. There is limited literature on the quality of integrated reports, and the IIRC itself does not define quality in reporting (Atkins et al., 2020; Marx & Mohammadali-Haji, 2014). However, it has provided detailed fundamental concepts, guiding principles and content elements which, when followed, result in a useful, fit-for-purpose report. Some authors have attempted to summarise features of high-quality reports based on technical reviews of South African integrated reports. Examples of some of the features identified in literature include stakeholder-centric reporting and an appropriate balance between positive and negative consequences of an organisation's business model on the different capitals (Atkins et al., 2020).

Within the integrated reports, companies often rely on other sustainability frameworks, such as the GRI framework and the UN SDGs, to compile sustainability information (Setia et al., 2024). Siwela (2025) explained that the JSE has developed Sustainability Disclosure Guidance, along with a Climate Change Disclosure Guidance specifically tailored to the South African context. While intended primarily to assist JSE-listed companies, this guidance is envisaged to be of value to institutional investors and the different entities that they invest in (including non-listed companies and debt issuers), as well as a range of stakeholder groups interested in sustainability/ESG disclosure and performance (JSE, 2024).

Overall, the extant literature on sustainability reporting and corporate governance presents mixed and inconclusive findings. While several studies document positive associations between board characteristics, such as size, diversity, and expertise, and sustainability reporting (Amran et al., 2014; Correa-Garcia et al., 2020; Kiliç et al., 2015; Masud et al., 2018), others report insignificant or conflicting results (Haladu & Bin-Nashwan, 2022; Khan, 2010). These inconsistencies suggest that the relationship between governance structures and sustainability outcomes may be context-dependent and influenced by institutional and regulatory environments. Furthermore, much of the existing literature has focused on developed markets and on quantitative firm characteristics, with limited attention given to qualitative governance attributes within emerging market settings (Al-Qudah & Houcine, 2024; Shamil et al., 2014). In addition, prior studies often examine individual board characteristics in isolation, with less emphasis on their combined influence on SRQ. This study, therefore, seeks to address these gaps by examining the influence of multiple board governance characteristics on SRQ within the South African context, an advanced yet underexplored integrated reporting environment.

In general, many sustainability reporting guidance tools used by companies on the JSE have strong links with and rely on the most recognised global initiatives on sustainability and climate change disclosure, such as the GRI Sustainability Reporting Standards and the Task Force on Climate-related Financial Disclosures (TCFD). These widely endorsed frameworks specify the essential traits of high-quality sustainability reporting. For instance, according to Cappellieri et al. (2025), the GRI principles put forward six key attributes that, when followed, should lead to high-quality sustainability reporting as shown in Table 1. For Cappellieri et al. (2025), these attributes include balance, comparability, accuracy, timeliness, clarity and reliability. Consequently, several scoring systems which have been developed to measure SRQ have been based on these attributes.

Given these mixed findings, further empirical investigation is required to better understand how board characteristics influence SRQ in emerging market contexts.

2.2 Hypothesis Development

2.2.1 Board size

The literature presents mixed evidence on the relationship between board size and SRQ. The first view holds that larger boards exhibit greater diversity in terms of experience, financial capabilities and problem-solving abilities. Some authors posit that the diversity in larger boards represents wider interests and takes advantage of human and social capital, which improves the quality of board decisions (Amran et al., 2014). Further strengthening this view, Chang et al. (2017) suggested that a larger board is likely to facilitate better disclosure of financial, social and environmental information, hence reducing risks and uncertainties.

The other view argues that larger boards suffer from inefficiency and poor coordination problems, which lead to weaker management control and increased agency costs (Girella et al., 2021). There is an argument that the lack of communication and coordination results in a lack of unanimity and director independence, hence reducing the quality of financial disclosures (Amran et al., 2014; Kiliç et al., 2015).

Masud et al. (2018) examined the effect of corporate governance elements on environmental sustainability performance in South Asian countries. The results found a positive association between board size and environmental sustainability reporting performance. Correa-Garcia et al. (2020) examined the factors impacting SRQ in non-financial groups from Latin American countries from 2011 to 2015. Their results revealed a positive

and statistically significant relationship between larger boards and SRQ.

Table 1. The GRI guiding principles definitions for quality

Principle	Explanation
Balance	The report should include both negative and positive information regarding an entity's ESG performance and practices.
Comparability	The matters identified should be reported on consistently and presented in a manner that allows comparability and analysis over time.
Accuracy	The entity must present information that is comprehensive, accurate and verifiable. It is best practice to have a third-party assurer.
Timeliness	The entity must report on ESG issues regularly and timeously to allow stakeholders to make sound decisions.
Clarity	The report should present information in a manner that is both understandable and easily accessible to stakeholders.
Reliability	The information and processes involved in preparing the sustainability report must be systematically gathered, recorded, compiled, analysed, and disclosed in a way that ensures their verifiability and confirms the quality and materiality of the information.

Note: GRI, Global Reporting Initiative; ESG, environmental, social, and governance. Source: Cappellieri et al. (2025).

Other studies have found no association between board size and sustainability reporting. For instance, Haladu & Bin-Nashwan (2022) examined the extent to which environmental agency oversight moderates the relationship between firm attributes, including board size, and sustainability reporting. Their results revealed that board size had no significant association with SRQ. Based on the literature review, it is hypothesised that:

H1: There is a positive association between board size and SRQ.

2.2.2 Gender diversity

Discussions around board gender diversity in corporate governance research have gained prominence in recent years (Adams & Ferreira, 2009; Williams, 2003). The agency theory has been used to link board diversity to improved monitoring, enhanced decision making and better understanding of complexities surrounding the modern firm (Carter et al., 2003; Hillman et al., 2009). The resource dependency theory has also been used to emphasise the role of gender diversity in connecting with external resources (Pfeffer & Salancik, 2003). Board gender diversity has been identified as a strategic resource that links the internal business environment to the external environment, hence, establishing a competitive advantage (Gallego-Álvarez et al., 2010).

Bear et al. (2010) pointed out that, from a legitimacy perspective, it is considered that female directors are more likely to advocate for community and social responsibility projects. Furthermore, female directors on the board are perceived to encourage stakeholder engagement and enhance the credibility of corporate reports (Manetti & Toccafondi, 2012; Ntim & Soobaroyen, 2013). These actions are likely to increase the legitimacy of the firm amongst stakeholders.

A study by Al-Shaer & Zaman (2016) examined the link between board gender diversity and SRQ. They used five different measures for board gender diversity on a sample of companies listed in the UK Financial Times-Stock Exchange 350 Index (FTSE 350) in 2012. Interestingly, their results revealed a positive association between gender diversity and SRQ.

These results are supported by Singhanian et al. (2024), who examined the role of gender diversity in sustainability disclosures. Their findings suggest that an increase in the percentage of female directors sitting on the board will improve SRQ.

Some studies have neither found statistical significance nor a relationship between board gender diversity and SRQ. Amran et al. (2014) examined the role of the board of directors in SRQ on a sample of 113 companies across 12 countries in the Asia-Pacific and found no relationship between board gender diversity and SRQ. These results are similar to those found by Khan (2010) in an earlier study conducted on Bangladeshi commercial banks.

Overall, the literature presents mixed evidence regarding the relationship between gender diversity and SRQ, with some studies reporting positive associations while others find insignificant relationships. These inconsistencies suggest that the influence of gender diversity may be context-specific and shaped by differences in governance and institutional environments.

Based on the reviewed literature, it is hypothesised that:

H2: There is a positive association between gender diversity and SRQ.

2.2.3 Ethnic diversity

The relationship between ethnic diversity and sustainability disclosures is underexplored in the literature (Shamil et al., 2014). The agency theory and resource dependence theory perspectives posit that diversity enhances the level of scrutiny through the consideration of diverse perspectives (Ntim, 2015). Furthermore, diverse boards are considered to represent the interests of a wider group of stakeholders, which aligns the organisation with the

various needs of stakeholders. Consequently, ethnically diverse boards are likely to demonstrate greater commitment towards sustainability practices, including reporting for a wide range of stakeholders (Ayuso & Argandoña, 2009; Galbreath, 2016).

Empirical literature is inconclusive on the influence of ethnic diversity on SRQ. Some studies have reported that multi-ethnic boards have a positive influence on sustainability issues and corporate social responsibility (CSR) corporate disclosures (Ntim & Soobaroyen, 2013; Zhang, 2012). However, others have rejected a link between ethnic diversity and sustainability disclosures (Shamil et al., 2014).

Overall, the limited literature on ethnic diversity and sustainability reporting presents inconclusive findings, highlighting the need for further evidence from emerging market contexts, particularly within governance environments characterised by transformation and inclusivity reforms.

Based on the reviewed literature, it is hypothesised that:

H3: There is a positive association between ethnic diversity and SRQ.

2.2.4 Board age

The broad effects of the age of the board on the quality of sustainability reporting are unexplored. However, the importance of the age of the board of directors in governance matters is recognised in the resource dependency theory (Katmon et al., 2019). Against this background, others have explored the link between the age of the board and CSR and related disclosures (Post et al., 2011; Songini et al., 2022).

Ararat et al. (2010) highlighted that differences in the ages of board members reflect the differences in their values and experiences, which leads to wider perspectives amongst the board. Mahadeo et al. (2012) explained the contributions brought by the three age categories, namely, elderly members, middle-aged members, and younger members. They posit that older board members contribute through their vast experience, expansive networks, and financial resources, while middle-aged members perform executive tasks, and the younger directors gain knowledge and experience of the organisation.

In another view, it is the younger directors who take more risks and display a keenness to display more on the entity's CSR, while older directors are more apt to err on the side of caution by engaging in less risky behaviour and minimising CSR disclosure (Katmon et al., 2019). Bekiroglu et al. (2011) added to this view by suggesting that younger directors are more likely to immerse themselves in ethical and environmental issues.

Songini et al. (2022) examined the impacts of board characteristics on corporate reporting quality, specifically integrated reporting quality. They studied a sample of 53 firms from various regions of the world, including Europe and Africa. Although their results showed that the age of the board of directors can impact SRQ, they did not support their hypothesis, which stated that the quality of integrated reporting would improve with the decrease in average age of the board.

Based on the review of the limited extant literature on this relationship, it is hypothesised that:

H4: There is an association between the average age of the board and the quality of sustainability reporting.

2.2.5 Board independence

Many corporate governance reforms, including the King IV code, recommend that the board of directors should be composed of a larger number of Non-Executive Directors (NEDs), most of whom should be independent (Institute of Directors South Africa, 2016). Independent directors or 'outside directors' do not perform any executive functions within an organisation and are therefore not involved in the day-to-day operations of the entity (Bursa Malaysia Securities Berhad, 2013). The extant literature employs the agency theory to support the view that increasing the number of independent directors can assist in mitigating agency costs and enhancing the quality of the monitoring role of the board (Fama & Jensen, 1983; Jensen & Meckling, 1976).

Kesner & Johnson (1990) highlighted the resource dependency theory perspective, which encourages the inclusion of more independent directors on the board, due to the expertise, networks, and prestige they offer. On the other hand, others argue that independent directors do not possess enough knowledge of the organisation's operations, especially those who have not served on the board for a long tenure. In this regard, they posit that strategy setting and performance monitoring are likely to be less effective (Carter & Lorsch, 2004; Jamil et al., 2021). Amran et al. (2014) explained that independent directors have a strong association with external stakeholders. Moreover, there is an increased expectation that they will protect the interests of the organisation and its stakeholders. In this context, they are more likely to request more information and advocate for the disclosure of high-quality information and promote informed decision-making for the organisation and its stakeholders (Vitolla et al., 2020).

Vitolla et al. (2020) examined the effect of board independence and integrated reporting quality. They found a positive association between board independence and integrated reporting quality. These results are consistent with Masud et al. (2018), who studied the influence of corporate governance on environmental sustainability reporting on 88 listed firms across three different countries in the South Asian region. Their results revealed a positive association between board independence and environmental sustainability reporting.

Based on the review of literature, it is hypothesised that:

H5: There is a positive association between board independence and SRQ.

2.2.6 Financial expertise

Literature represents broad financial expertise as the percentage of board members with accounting and finance-related qualifications and expertise (Erin et al., 2022; Githaiga & Kosgei, 2023). Board reliance on industry experience alone is regarded by some authors as insufficient to adequately address the broad scope of issues in corporate governance (Umukoro et al., 2019).

Directors with an accounting and finance background possess a better understanding of financial reports. Hence, they are more equipped to make sound financial decisions for the firm (Erin et al., 2022). Furthermore, their training on social accounting enables them to have a better understanding of ESG issues. Against this background, directors with financial expertise are more likely to advocate for sustainability reporting (Ahmad et al., 2018).

Ahmad et al. (2018) examined the association between board characteristics and CSR reporting in Malaysian companies. Their results revealed a significant and positive relationship between board financial expertise and CSR reporting. Erin & Adegboye (2022) examined the impact of board characteristics and integrated reporting quality. Their results revealed a positive and significant association between board financial expertise and integrated reporting quality.

Based on the literature review, it is hypothesised that:

H6: There is a positive association between board financial expertise and SRQ.

3. Research Method

The study examines the relationship between corporate governance and an integrated-report-based proxy for SRQ in 37 firms drawn from the JSE Top 40. The JSE Top 40 Companies represent approximately 80% of the total market capitalisation, and is widely used in South African corporate governance research due to its coverage (van Zijl & Hewlett, 2022).

While the Top 40 index provides extensive coverage of market capitalisation and captures most of the economic activity on the JSE, it is important to note that this sample is skewed towards large, well-established firms. Consequently, the governance structures, resource availability, and reporting practices observed in this study are likely to differ materially from those of small and medium-sized enterprises (SMEs), which typically operate under more constrained financial and organisational conditions. As such, the findings of this study should be interpreted within the context of large, listed firms and not as representative of the broader population of JSE-listed or unlisted entities.

The study adopted a purposive convenience sampling strategy due to the ease of obtaining data. Two inclusion criteria were applied in determining the final sample: (1) the company had a primary listing on the JSE and published an integrated report in every year of the 2019 to 2023 period; and (2) the company received an Ernst and Young (EY) disclosure quality rating in every year under study.

Diversity variables, which are employed by the study as independent variables, are emphasised in the King IV report of 2016. However, many companies on the JSE would have integrated these recommendations in the years that followed. The time frame is therefore considered ideal to capture the long-term impact of the King IV report recommendations on sustainability reporting. Furthermore, this time frame captures the impact of the COVID-19 pandemic on corporate governance and ESG factors. Several companies were forced to re-evaluate their governance structures and strengthen board oversight on ESG issues (Adams & Abhayawansa, 2022). Analysing the data from 2019–2023 would capture the impact of the most recent governance shifts on an integrated-report-based proxy for SRQ. Qualitative and quantitative data were collected for the 2019–2023 period for 37 companies with the highest market capitalisation.

3.1 Measurement of Variables

3.1.1 Independent and control variables

To account for firm-level heterogeneity and to estimate the conditional association between board composition and SRQ, several control variables were included on the basis of their documented relationship with SRQ (Bhatia & Tuli, 2017; Nursimloo et al., 2020; Shamil et al., 2014). Table 2 below shows the measurement of independent and control variables in line with previous studies (Githaiga & Kosgei, 2023; Tahir et al., 2023).

3.1.2 Dependent variable

The study seeks to examine the association of corporate governance variables on SRQ. In this instance, SRQ was measured using the EY Excellence in Integrated Reporting Awards as a proxy. This became the dependent variable. Scoring systems are prominent in measuring corporate reporting quality (Amran et al., 2014; Correa-Garcia et al., 2020; Erin et al., 2022; Pistoni et al., 2018). These scoring systems have been constructed using criteria outlined by sustainability reporting frameworks as benchmarks for reporting quality and assumptions of

suitable measures of corporate reporting quality.

Table 2. Variable definition

Variable	Definition
Board size	Natural logarithm of total board size
Board independence	Proportion (Percentage) of board members who are independent NEDs
Average age of the board	Natural logarithm of the average age of board members
Gender diversity	Proportion (Percentage) of board members who are females
Financial expertise	Proportion (Percentage) of board members with accounting and financial expertise
Control Variables	
Firm performance/ Profitability	Return on assets
Firm size	Logarithm of total assets
Firm age	Logarithm of the number of years since the firm was incorporated

Note: NEDs, Non-Executive Directors.

The EY Excellence in Integrated Reporting framework, while grounded in IIRC principles, reflects broader SRQ constructs that overlap with widely recognised frameworks such as the GRI and TCFD. Core attributes such as balance, reliability, completeness and forward-looking risk disclosures are consistent with these global standards. However, the EY ratings remain context-specific to South Africa and are not directly comparable to international assessment systems, as no universally standardised external rating framework currently exists.

The study used scores from the EY Excellence in Integrated Reporting Awards, a widely acknowledged and accepted disclosure rating system, as a proxy for reporting quality (Barth et al., 2017; Eloff & Steenkamp, 2022). EY assesses the quality of integrated reporting disclosures for the Top 100 JSE-listed companies. EY's scoring system is based on the IIRC's content elements and guiding principles, which focus on long-term value creation. Naturally, this includes a wide range of sustainability and ESG-related disclosures. Additionally, Cortesi & Vena (2019) expressed strong links between integrated reporting quality and sustainability-related performance and disclosure quality (Mans-Kemp & Van der Lugt, 2020). EY itself, in its Excellence in Integrated Reporting Award, proffers a correlation between integrated reporting quality and sustainability reporting effectiveness (EY, 2023).

EY's scoring system categorises the quality of reporting into four key areas, namely: 1) 'Excellent', 2) 'Good', 3) 'Average', and 4) 'Progress to be made'. For the purposes of the analysis, the original EY Excellence in Integrated Reporting Awards classifications were applied, resulting in four outcome categories: Progress to be made, Average, Good, and Excellent, which justified the use of multinomial logistic regression. It has been explained by Herbert (2018) that not all JSE-listed companies produce a stand-alone sustainability report; however, all JSE-listed companies with a primary listing issue a form of integrated report, which includes company sustainability disclosures. Against this background, assessing the quality of sustainability reporting within the integrated reports would allow the study to capture wider insights on sustainability reporting practices on the JSE. Furthermore, the metrics used to assess integrated reporting quality, such as completeness, balance, comparability and reliability, are key elements of SRQ as defined by several sustainability reporting frameworks, including the GRI guidelines on SRQ (EY, 2023; GRI, 2013; Safari & Areeb, 2020).

The assessment is conducted using a structured rubric that evaluates reports across several qualitative dimensions, including completeness, balance, conciseness, and the extent to which organisations demonstrate integrated thinking. As a result, the proxy used in this study reflects a context-specific measure of reporting quality that is closely aligned with South Africa's advanced integrated reporting environment but may not be directly comparable to international sustainability reporting assessments.

Notwithstanding the structured nature of this framework, the evaluation process inherently involves a degree of professional judgement. To mitigate subjectivity, EY applies a panel-based review process involving multiple assessors and standardised evaluation criteria. However, as with most disclosure quality indices, some degree of evaluator discretion remains unavoidable. This limitation should be considered when interpreting the results, particularly in comparison to purely quantitative scoring approaches.

The advantage of employing the EY rating system lies in the use of an established and nationally recognised rating system, external to the study. This is important in reducing subjective bias caused by an author-developed scoring system and promoting the reliability and validity of the results (Mans-Kemp & Van der Lugt, 2020). The limitations of EY's methodology were also considered. Because the ratings assess integrated reports as a whole, they capture reporting quality broadly across financial, governance and sustainability dimensions, rather than providing a depth of coverage specific to sustainability disclosures. This constraint was accepted on the basis that an established external rating offers greater objectivity than an author-developed instrument, and the resulting findings are accordingly interpreted within the context of integrated reporting.

3.2 Model Specification

The study aims to understand the association of board governance characteristics with SRQ using 37 JSE listed firms. SRQ, measured using the EY Excellence in Integrated Reporting Awards, was modelled as a function of the board size, structure and firm-specific characteristics shown in Eq. (1). The model specification is shown below:

$$SRQ = f(\text{board structure and firm specific characteristics}) \quad (1)$$

The board structure consists of board size, percentage of women in the board, percentage of people of colour in the board (black, Indian or coloured), independent members in the board, age of board members and the inclusion of an expert in finance in the board. The firm's specific characteristics include firm size, profitability and age. Eq. (2) illustrates the panel regression equation as follows:

$$y_{it} = \beta_0 + \sum_i^n x_{it}\beta_i + \sum_i^n z_{it}\beta_i + \varepsilon_{it} \quad (2)$$

where, y_{it} is sustainability quality reporting for the company i at time t in years. Independent variables signifying board structure are represented by x_{it} and control variables are denoted by z_{it} . The unknown parameters are β_0 and β_i including an error term ε_{it} . Eq. (3) illustrates the model of this study as shown below:

$$SRQ_{it} = \beta_0 + \beta_1 IBS_{it} + \beta_2 GWD_{it} + \beta_3 EAD_{it} + \beta_4 IAAD_{it} + \beta_5 IND_{it} + \beta_6 FE_{it} + \beta_7 IFS_{it} + \beta_8 P_{it} + \beta_9 IFA_{it} + \varepsilon_{it} \quad (3)$$

where, SRQ represents EY Excellence in Integrated Reporting Awards proxy for sustainability reporting quality, which is the dependent variable having four levels: Progress needs to be made, Average, Good and Excellent. The explanatory variables are IBS—log board size, GWD—gender diversity focusing on women, EAD—ethnic diversity focusing on African members, FE—financial expertise, IAAD—log average age of board, IND— independent members, P—profitability, IFA—log firm age, and IFS—log firm size.

3.3 Model Specification

The multinomial logistic regression model is an extension of binary logistic regression used for multi-class classification problems. While binary logistic regression is applied when the dependent variable has two outcomes, the multinomial approach allows for more than two categories. In this study, the dependent variable, SRQ, is classified into four categories: 'Progress needs to be made', 'Average', 'Good', and 'Excellent'. The model estimates the log-odds of each outcome category relative to a reference category as a function of the independent and control variables. These estimated logits are then used to derive the predicted probabilities of each class. Accordingly, the coefficients are interpreted in terms of their effect on the log-odds of belonging to a particular reporting quality category rather than as direct changes in probabilities.

Multinomial logistic regression is used as an expansion of the logistic model as more than one independent variable is applied in the model. Let k denote number of independent variables for a binary dependent Y by $x_1, x_2 \dots x_k$, the log odds model shown below:

$$\text{Logit}[P(Y = 1)] = \alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k \quad (4)$$

The parameter β_i represents x_i effect on the log odds of $Y = 1$ and an intercept of α . This can then be applied to our SRQ levels under a multinomial logistic model using firm-level clustered standard errors, as shown in the equation below:

$$\ln\left(\frac{P(SRQ_{it} = j)}{P(SRQ_{it} = 0)}\right) = \alpha_j + \beta_{1j} IBS_{it} + \beta_{2j} GWD_{it} + \beta_{3j} EAD_{it} + \beta_{4j} IAAD_{it} + \beta_{5j} IND_{it} + \beta_{6j} FE_{it} + \beta_{7j} IFS_{it} + \beta_{8j} P_{it} + \beta_{9j} IFA_{it} + \varepsilon_{it} \quad (5)$$

where, SRQ_{it} is sustainability reporting quality for firm i in year t . The base category is 'Progress to be made'. The model is estimated as a multinomial logit, but the standard errors are clustered at the firm level to correct for repeated observations within the same firm over time. The coefficients α_j and β_j are unknown parameters whilst the ε_{it} is the stochastic error term assumed to be identically distributed and independent with a zero mean and a variance that is constant.

4. Data Analysis and Findings

Data collected from both the Equity RT database and official company websites spanned across the years 2019 to 2023 and included 37 companies, 10 variables, and 185 observations for panel data analysis. Data was collected and processed, that is other variables converted to logarithmic values and other as percentages of board size. Pre-tests and diagnostic tests were conducted like unit root tests, correlation analysis, multicollinearity, autocorrelation, heteroskedasticity and normality.

4.1 Data Analysis

Two well-known and flexible platforms, Stata 18 and R software 4.3.3, were employed in this study to investigate the connection between board composition and SRQ. Stata is an excellent tool for performing complex statistical analyses like multinomial logistic regression because of its user-friendly interface and wide range of econometric functions (Mehmetoglu & Jakobsen, 2022). Conversely, R provides a flexible programming environment with sophisticated features for model validation, data manipulation, and visualization (Wiley & Wiley, 2019).

Based on the descriptive statistics in Table 3, the data reveals significant insights into the dataset. The variable board size (IBS) exhibits a mean of 1.057, with a median of 1.079, indicating a slight skewness towards lower values, as the mean is less than the median. The gender diversity (GWD) variable shows a substantial range, with a minimum of 0 and a maximum of 100, and a mean of 36.31, suggesting a wide dispersion in the data. Similarly, ethnic diversity (EAD) has a mean of 28.78 and a median of 30.77, with values ranging from 0 to 83.33, indicating variability in the dataset.

Table 3. Summary statistics

Variable	Min	1st Quartile	Median	Mean	3rd Quartile	Max
IBS	0.602	1.000	1.079	1.057	1.146	1.301
GWD	0.000	26.670	30.770	36.310	35.710	100.000
EAD	0.000	12.500	30.770	28.780	40.000	83.330
FE	25.000	30.770	38.460	43.850	53.850	100.000
IAAD	1.635	1.740	1.756	1.762	1.778	1.886
IND	30.770	66.670	71.430	71.790	76.920	107.140
P	-7.210	1.230	4.720	7.127	9.430	52.490
IFS	5.129	7.908	8.179	8.142	8.669	10.711

Note: IBS, log board size; GWD, gender diversity focusing on women; EAD, ethnic diversity focusing on African members; FE, financial expertise; IAAD, log average age of board; IND, independent members; P, profitability; IFS, log firm size.

The financial expertise (FE) variable, with a mean of 43.85 and a median of 38.46, shows a higher concentration of values in the upper quartiles. Average age of board (IAAD) variable presents a relatively narrow range, with a mean of 1.762 and a median of 1.756, suggesting consistency in the data. The independent members (IND) variable, with a mean of 71.79 and a median of 71.43, indicates a balanced distribution around the central tendency. The profitability (P) variable shows a significant range from -7.210 to 52.490, with a mean of 7.127, highlighting potential outliers. Lastly, the firm size (IFS) variable has a mean of 8.142 and a median of 8.179, indicating a symmetric distribution. Overall, the dataset exhibits a mix of variability and consistency across different variables, providing a comprehensive overview of the underlying data characteristics.

Table 4. Variance inflation factor (VIF) results

Variable	VIF Value
IBS	1.750
GWD	1.142
EAD	1.469
FE	1.319
IAAD	1.610
IND	1.278
P	1.040
IFS	1.089
IFA	1.084

Note: IBS, log board size; GWD, gender diversity focusing on women; EAD, ethnic diversity focusing on African members; FE, financial expertise; IAAD, log average age of board; IND, independent members; P, profitability; IFS, log firm size, IFA, log firm age.

4.2 Variance Inflation Factor Diagnostics

Table 4 illustrates the results of the variance inflation factor (VIF) test which tests multicollinearity. The rule of VIF analysis is that values above 5 indicate multicollinearity and the variables should be excluded from the regression analysis. In this study the variables had VIF values less than 5 indicating that no problematic multicollinearity was detected under the selected threshold.

4.3 Heteroskedasticity Tests

The Breusch-Pagan (BP) test was used to test Heteroskedasticity and the null hypothesis states that Heteroskedasticity does not exist. The test is conducted by checking the BP value and *p*-value.

The results in Table 5 show that the *p*-value is 0.117 which is greater than 0.05. Therefore, we fail to reject the null hypothesis of homoskedasticity, suggesting that there is no evidence of heteroskedasticity in the model.

Table 5. Heteroskedasticity tests results

Breusch-Pagan	<i>p</i> -Value
14.144	0.117

4.4 Multinomial Model Results Analysis and Interpretation

The multinomial model was applied to analyse the influence of board governance characteristics in sustainability reporting using 37 listed firms. The results obtained from the Multinomial logit model with firm-level clustered standard errors are shown in Table 6.

Table 6. The multinomial logit with firm-level clustered standard errors

SRQ Category	Variable	Coefficient (β)	Odds Ratio (e^{β})	Clustered SE	z-Value	<i>p</i> -Value
Average	IBS	-5.134	0.006	3.306	-1.553	0.120
Average	GWD	-0.017	0.983	0.012	-1.450	0.147
Average	EAD	0.036	1.037	0.025	1.426	0.154
Average	FE	-0.017	0.983	0.031	-0.565	0.572
Average	IAAD	7.657	2,116.35	9.758	0.785	0.433
Average	IND	0.013	1.013	0.040	0.333	0.739
Average	P	0.100	1.105	0.064	1.576	0.115
Average	IFS	-0.379	0.685	0.630	-0.602	0.547
Average	IFA	1.276	3.582	1.887	0.676	0.499
Good	IBS	-6.574	0.001	4.252	-1.546	0.122
Good	GWD	-0.011	0.989	0.008	-1.281	0.200
Good	EAD	0.050	1.051	0.032	1.552	0.121
Good	FE	-0.025	0.975	0.034	-0.725	0.468
Good	IAAD	2.713	15.07	10.826	0.251	0.802
Good	IND	-0.034	0.967	0.035	-0.969	0.333
Good	P	0.117	1.124	0.067	1.740	0.082*
Good	IFS	0.015	1.015	0.595	0.026	0.980
Good	IFA	0.025	1.025	1.592	0.016	0.987
Excellent	IBS	-6.742	0.001	4.776	-1.411	0.158
Excellent	GWD	-0.005	0.995	0.006	-0.819	0.413
Excellent	EAD	0.084	1.088	0.031	2.721	0.007***
Excellent	FE	-0.047	0.954	0.038	-1.238	0.216
Excellent	IAAD	6.922	1,014.44	10.593	0.653	0.513
Excellent	IND	-0.014	0.986	0.040	-0.346	0.730
Excellent	P	0.126	1.134	0.069	1.831	0.067*
Excellent	IFS	0.360	1.433	0.684	0.526	0.599
Excellent	IFA	-0.397	0.672	1.608	-0.247	0.805

Note: SRQ, sustainability reporting quality; IBS, log board size; GWD, gender diversity focusing on women; EAD, ethnic diversity focusing on African members; FE, financial expertise; IAAD, log average age of board; IND, independent members; P, profitability; IFS, log firm size; IFA, log firm age. Two-tailed tests, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6 shows the results of the multinomial Logit with firm-level clustered standard errors, illustrating the sign

of the coefficients, *p*-values and the decision on statistical significance of the independent variables.

4.5 Analysis of Findings

H1: There is a positive association between board size and SRQ.

The results revealed a negative association between board size and SRQ; however, this relationship was not statistically significant after correcting for firm-level clustering. In the context of the multinomial logit model, this indicates that an increase in board size is associated with lower log-odds and lower odds of a firm being classified into higher SRQ categories relative to the base category. The odds ratios for board size are substantially below one across all categories, suggesting lower odds of achieving higher SRQ classifications among firms with larger boards. Although the direction of the relationship is consistent with literature suggesting that larger boards may be associated with coordination challenges, slower decision-making processes, and reduced monitoring effectiveness (Amran et al., 2014; Girella et al., 2021), the evidence is insufficient to conclude that board size significantly influences SRQ. Within the sample of large, complex organisations, the observed negative association may be consistent with governance environments in which such challenges are more likely to arise. However, the present study does not directly test these mechanisms and therefore cannot establish whether they explain the relationship. In contrast, smaller firms may operate under different governance structures characterised by more streamlined decision-making processes. Therefore, the hypothesis is rejected.

H2: There is a positive association between gender diversity and SRQ.

Contrary to the hypothesis, the study observed a negative association between gender diversity and SRQ; however, this relationship was not statistically significant. This implies that changes in gender diversity were not significantly associated with either the log-odds or odds of belonging to higher sustainability reporting categories. The estimated odds ratios are slightly below one, indicating a marginal reduction in the odds of achieving higher SRQ classifications as gender diversity changes, although the relationship remains statistically insignificant. These findings are consistent with Amran et al. (2014), who also found no significant relationship, but contrast with Al-Shaer & Zaman (2016), who reported a positive association in developed markets. The results suggest that gender diversity alone may not be sufficient to be related to sustainability reporting outcomes, and that its association may depend on broader governance and institutional factors. Therefore, the hypothesis is rejected.

H3: There is a positive association between the ethnic diversity of the board and SRQ.

The results showed a positive association between ethnic diversity and SRQ, with the relationship remaining statistically significant for firms classified in the highest SRQ category after correcting for firm-level clustering. In multinomial terms, this indicates that increased ethnic diversity is associated with higher log-odds and odds of firms being classified into higher SRQ categories relative to the base outcome. The odds ratio for the Excellent category is 1.088, implying that a one-unit increase in ethnic diversity is associated with higher odds of achieving Excellent SRQ by approximately 8.8%, holding other variables constant. This finding aligns with literature which suggests that ethnically diverse boards represent a broader range of stakeholder interests and enhance decision-making through diverse perspectives (Ntim & Soobaroyen, 2013; Zhang, 2012). In the South African context, where governance reforms emphasise inclusivity and transformation, ethnic diversity is positively associated with higher SRQ and may be linked to stronger sustainability disclosures (Bezuidenhout et al., 2007; Formanek, 2021). However, these results reflect statistical associations rather than causal mechanisms. Therefore, the hypothesis is accepted.

H4: There is an association between the average age of the board and SRQ.

The results show a positive association between average board age and SRQ; however, the relationship was not statistically significant after adjusting for firm-level clustered standard errors. This suggests that an increase in board age was associated with higher estimated log-odds and odds of achieving higher SRQ classifications, although the evidence is insufficient to establish a statistically reliable relationship. While the estimated odds ratios are greater than one, indicating higher odds of superior reporting quality, the large standard errors and lack of statistical significance require cautious interpretation. These findings contribute to the limited literature on board age and sustainability reporting and remain broadly consistent with arguments that older board members bring valuable experience, networks, and strategic insights (Jonson et al., 2020; Katmon et al., 2019). While some studies suggest younger boards may be more sustainability-oriented, the current findings indicate that experience and institutional knowledge may be associated with higher reporting quality within this sample. Therefore, the hypothesis is not supported.

H5: There is a positive association between board independence and SRQ.

The results reveal a mixed but statistically insignificant relationship between board independence and SRQ. While a positive association is observed for the 'Average' category and negative associations for the 'Good' and 'Excellent' categories, these effects are not statistically significant, implying that board independence was not statistically associated with either the log-odds or odds of achieving higher SRQ levels. The corresponding odds ratios are close to one across categories, suggesting limited practical influence on SRQ. This suggests that increasing the number of independent directors may not necessarily translate into improved sustainability oversight.

The findings may reflect “symbolic independence,” where firms expand board size to meet governance requirements without enhancing functional effectiveness. Additionally, the absence of ESG-specific expertise among independent directors may limit their influence on sustainability reporting. Therefore, the hypothesis is rejected.

H6: There is a positive association between board financial expertise and SRQ.

The results indicate a negative and statistically insignificant relationship between financial expertise and SRQ after correcting for within-firm correlation. This suggests that financial expertise was not significantly associated with either the log-odds or odds of belonging to higher SRQ categories. The estimated odds ratios are below one across the reporting categories, indicating a reduction in the odds of achieving higher SRQ classifications, although the relationship remains statistically insignificant. The direction of the relationship contrasts with prior studies (Ahmad et al., 2018; Erin et al., 2022), which report positive associations. One possible explanation is that financially skilled directors may place relatively greater emphasis on financial performance and compliance than on broader ESG disclosures. A further hypothetical explanation, not tested here, is that financial expertise may emphasise quantifiable metrics, which could be associated with conservative approaches toward qualitative sustainability reporting. Overall, financial expertise does not appear to play a decisive role in shaping SRQ in this sample. Therefore, the hypothesis is rejected.

5. Discussion

The relationship between board governance characteristics and an integrated-report-based proxy for SRQ was analysed using a multinomial logistic regression model estimated with firm-level clustered standard errors, where the estimated coefficients represent the log-odds of a firm being classified into a specific SRQ category relative to the base category (‘Progress to be made’). To improve interpretation, odds ratios were also computed by exponentiating the estimated coefficients. The coefficients therefore indicate the direction and strength of the association with the likelihood of belonging to a higher SRQ category relative to the base outcome, while the odds ratios indicate the magnitude of the association after accounting for repeated observations within firms over time.

The findings indicate that board size (IBS) has a negative association across all SRQ categories; however, the relationship is not statistically significant after correcting for firm-level clustering. This implies that larger board size is associated with lower log-odds and odds of achieving higher SRQ, although the evidence is insufficient to conclude that board size is a significant determinant of reporting quality. The odds ratios for board size are substantially below one across all categories, indicating lower odds of firms being classified into higher SRQ categories as board size increases. The direction of the relationship remains consistent with arguments in the literature suggesting that larger boards may be associated with coordination challenges, slower decision-making processes, and weaker monitoring effectiveness (Amran et al., 2014; Girella et al., 2021). Within the South African context, where governance structures are already complex, the observed negative association may be consistent with governance environments in which such challenges are more likely to occur. However, the present study does not directly examine these mechanisms and therefore cannot establish whether they explain the relationship between board size and SRQ.

In contrast, ethnic diversity (EAD) demonstrates a positive relationship across all categories and remains statistically significant in the ‘Excellent’ category after correcting for firm-level clustering, suggesting that more ethnically diverse boards are associated with higher log-odds and odds of achieving higher SRQ classifications. The odds ratio of 1.088 in the ‘Excellent’ category indicates that a one-unit increase in ethnic diversity is associated with approximately 8.8% higher odds of achieving excellent SRQ, holding other variables constant. This aligns with prior studies which argue that diverse boards incorporate broader stakeholder perspectives and enhance decision-making related to sustainability disclosures (Ntim & Soobaroyen, 2013; Zhang, 2012). Given South Africa’s transformation-driven governance environment, this result further reflects the importance of inclusivity in strengthening corporate accountability and legitimacy (Bezuidenhout et al., 2007; Formanek, 2021).

The results further reveal that average board age (IAAD) shows a positive estimated association across SRQ categories; however, the relationship is not statistically significant after adjusting for firm-level clustered standard errors. This suggests that boards with greater experience and institutional knowledge may be statistically associated with higher SRQ, although the evidence is not sufficiently strong to establish a statistically reliable relationship. While the corresponding odds ratios are greater than one, indicating higher odds of superior SRQ, the large standard errors suggest substantial uncertainty around the estimates. These findings remain broadly consistent with the view that older directors contribute valuable expertise, networks, and strategic insights, which may be associated with stronger governance and disclosure practices (Katmon et al., 2019; Jonson et al., 2020).

With respect to financial expertise (FE), the findings show a negative and statistically insignificant relationship across the SRQ categories, indicating that higher levels of financial expertise are associated with lower odds of achieving superior SRQ, although the relationship is not statistically significant. The odds ratios are below one across categories, suggesting reduced odds of belonging to higher sustainability reporting categories. This result contrasts with prior studies (Ahmad et al., 2018; Erin et al., 2022). One possible explanation is that financially

skilled directors may place relatively greater emphasis on financial performance and compliance than on broader ESG disclosures. However, the present analysis does not directly test this explanation and therefore the relationship should be interpreted as an association rather than evidence of a causal mechanism.

The analysis also indicates that profitability (P) is positively associated with SRQ across all SRQ categories and remains marginally significant in the 'Good' and 'Excellent' categories after clustering. This implies that more profitable firms are associated with higher log-odds and odds of producing better-quality sustainability reports. The odds ratio of 1.124 for the 'Good' category and 1.134 for the 'Excellent' category suggests that a one-unit increase in profitability is associated with approximately 12.4% and 13.4% higher odds of being classified into these reporting categories, respectively. This finding is consistent with existing literature, which suggests that financially successful firms may have greater resources to invest in sustainability initiatives and reporting infrastructure (Ali et al., 2021; Hahn & Kühnen, 2013; Haniffa & Cooke, 2005; Herbert & Graham, 2022; Shamil et al., 2014). A further explanation, not tested in this study, is that profitable firms may be subject to greater stakeholder scrutiny, which may encourage higher-quality disclosures.

Regarding control variables, firm size (IFS) exhibits mixed effects, with a negative association in the 'Average' category and positive associations in the higher SRQ categories. However, these relationships are not statistically significant after correcting for within-firm dependence. While the direction of the coefficients and odds ratios suggests that larger firms may be more likely to achieve superior reporting quality, the evidence is insufficient to establish a robust relationship. Nevertheless, this finding remains broadly consistent with the argument that large firms have greater visibility, regulatory pressure, and access to resources necessary for comprehensive sustainability reporting (Hahn & Kühnen, 2013). These effects should be interpreted cautiously, as the sample consists primarily of large-cap firms.

Interestingly, firm age (IFA) shows inconsistent and statistically insignificant results across categories, suggesting that organisational maturity does not necessarily translate into improved SRQ. The odds ratios indicate positive effects in some categories and negative effects in others, reinforcing the absence of a clear and consistent relationship. This may indicate that younger firms are more adaptive and responsive to emerging ESG reporting practices, while older firms may rely on established reporting structures. However, the study does not directly examine these behavioural differences and therefore this interpretation should be treated as a possible explanation rather than a confirmed mechanism.

Importantly, the interpretation of these results should be grounded in the multinomial logit framework. For instance, a coefficient such as ($\beta_{IBS} = -5.134$) in the 'Average' category indicates that a one-unit increase in board size is associated with lower log-odds of a firm being classified as 'Average' relative to 'Progress to be made', holding other variables constant. The corresponding odds ratio of 0.006 indicates substantially lower odds of belonging to the 'Average' category relative to the base outcome. It does not imply a direct reduction in probability, as probabilities are non-linear transformations of the estimated logits. This distinction is critical for accurate interpretation and aligns with established econometric practice.

Overall, the findings suggest that board diversity, profitability, and board experience are more strongly associated with SRQ than board size. After accounting for within-firm correlation through clustered standard errors, ethnic diversity emerges as the governance characteristic most strongly statistically associated with superior SRQ, while profitability remains an important firm-level characteristic. The odds ratio results further demonstrate that ethnically diverse and profitable firms are more likely to achieve higher SRQ classifications. The results provide support for both agency theory, which emphasises monitoring efficiency and governance effectiveness (Fama & Jensen, 1983), and legitimacy theory, which highlights the role of stakeholder expectations and societal pressures in shaping corporate disclosures (Hahn & Kühnen, 2013; Herbert & Graham, 2022).

6. Conclusion and Recommendations

This study provides evidence on the relationship between board governance characteristics and an integrated-report-based proxy for SRQ within large, publicly listed firms in South Africa. The findings offer insights that are particularly relevant to large-cap companies operating within formalised governance structures, while caution should be exercised in extending these conclusions to smaller firms or SMEs.

6.1 Conclusion

Grounded mainly in agency and legitimacy theories, the present study examines the association between board characteristics and SRQ among listed firms in South Africa. The study included 37 firms listed on the JSE for 5 years and analysed 10 variables with 185 observations. The results indicate that ethnic diversity is statistically positively associated with SRQ, while average board age exhibits a positive but statistically insignificant relationship after correcting for firm-level clustering. Within the multinomial logit framework, this implies that increases in ethnic diversity are associated with higher log-odds and odds of firms being classified into superior

sustainability reporting categories relative to the base outcome. The odds ratio for the Excellent category indicates that a one-unit increase in ethnic diversity is associated with approximately 8.8% higher odds of achieving excellent SRQ. This is consistent with theoretical and empirical literature emphasising the value of diverse perspectives in enhancing strategic decision-making and disclosure quality. Additionally, the findings are consistent with the agency theory perspective, which underscores the importance of diverse viewpoints in improving governance outcomes. In line with previous empirical literature, the results also reveal a negative association between board size and SRQ, indicating that larger boards are associated with lower log-odds and odds of achieving higher reporting quality classifications, although this relationship is not statistically significant after clustering.

These results suggest that larger board size is associated with lower SRQ. While prior literature suggests that coordination challenges and slower decision-making may provide plausible explanations for this relationship, the present study does not directly test these mechanisms and therefore cannot establish causality. Interestingly, the results also show a negative and statistically insignificant relationship between financial expertise and SRQ. Interpreted within the multinomial framework, this suggests that higher levels of financial expertise are associated with lower odds of firms achieving top sustainability reporting classifications, although the evidence is insufficient to establish a statistically reliable relationship. One possible explanation is that financially oriented boards may place relatively greater emphasis on financial performance than on broader ESG considerations. However, the study does not directly examine this mechanism. Regarding the control variables, profitability demonstrates a positive relationship with SRQ and remains marginally significant for the Good and Excellent categories, suggesting that more profitable firms are associated with higher log-odds and odds of producing higher-quality sustainability reports. The odds ratios indicate that a one-unit increase in profitability is associated with approximately 12.4% and 13.4% higher odds of achieving Good and Excellent SRQ, respectively. This aligns with existing literature, which argues that financially successful firms have greater resources to invest in sustainability practices and reporting systems.

6.2 Recommendations

The results have important implications for policymakers, regulators, and theoretical development. For policymakers, understanding the determinants associated with effective board structures is essential in addressing stakeholder information needs. Preliminary evidence from this study suggests that attention may be given to board size, as larger boards were associated with lower SRQ within the sample examined. Policymakers may also consider broadening the scope of board expertise to include ESG-related competencies, as incorporating directors with sustainability expertise may strengthen the board's capacity to oversee sustainability reporting practices. Similarly, targeted sustainability training for financial experts on boards may be considered as a means of improving engagement with ESG-related disclosures. These implications are context-bound, drawing on a sample of 37 large JSE-listed firms. Interestingly, the study finds a statistically significant positive relationship between ethnic diversity and SRQ, while gender diversity remains statistically insignificant. This may reflect the stronger regulatory and institutional emphasis placed on ethnic transformation within the South African context. The positive association between ethnic diversity and SRQ suggests that more inclusive leadership structures are statistically associated with stronger sustainability reporting outcomes. Under the sample conditions examined in this study, no significant association was observed between gender diversity and SRQ. Whether such an association exists under different conditions and what factors might condition it remains an open question for further research and policy consideration rather than a basis for recommending regulatory intervention.

From a theoretical perspective, the findings are consistent with agency theory, which suggests that board effectiveness, monitoring quality, and diversity are important governance characteristics associated with improved organisational outcomes. The results also align with legitimacy theory, which anticipates that firms facing external pressure to conform to governance expectations will adjust their disclosure practices accordingly. One interpretation, which this study does not test, is that firms may expand board size in response to diversity requirements, producing symbolic compliance rather than substantive governance change. Taken together with the absence of a significant board size effect in this sample, the results suggest that board composition alone may be insufficient to explain variation in SRQ, although the board processes implied by this interpretation were not measured here and remain a question for further research.

6.3 Limitation

A limitation of the study is the use of the EY Excellence in Integrated Reporting ratings as an integrated-report-based proxy for SRQ. Although the EY ratings are widely recognised and grounded in the International Integrated Reporting Framework, they primarily evaluate the quality of integrated reporting rather than providing a direct and comprehensive measure of stand-alone SRQ. Consequently, the findings should be interpreted within the

context of integrated reporting practices rather than broader sustainability reporting frameworks such as GRI, Sustainability Accounting Standards Board (SASB), International Sustainability Standards Board (ISSB), or TCFD. Another limitation of this study is that, although the dataset has a panel structure, the analysis employs a multinomial logit model with firm-level clustered standard errors due to the short time dimension (2019–2023), which may not fully capture unobserved firm-specific effects over time. While the clustering approach adjusts for within-firm correlation and improves statistical inference, it does not explicitly model firm-specific heterogeneity.

A further limitation relates to the study's exclusive focus on the Top 40 JSE-listed companies. While these firms account for a substantial proportion of market capitalisation, they do not reflect the diversity of firm sizes within the South African economy. SMEs operate under different governance dynamics, ownership structures, and regulatory pressures, all of which may influence sustainability reporting practices in distinct ways. Consequently, the generalisability of the findings, especially those relating to board size, firm size, and profitability, is limited to large, listed firms. It is also important to note that the study does not capture internal organisational dynamics such as corporate culture or managerial attitudes, which may influence sustainability reporting practices. In addition, governance dynamics within SMEs are often less formalised and may involve greater managerial overlap, which could influence both board effectiveness and sustainability reporting practices differently from large, institutionalised firms. As such, extending these findings to SMEs would require careful consideration of these structural differences.

6.4 Future Studies

Future research could extend this work by applying panel multinomial techniques, such as random-effects multinomial models or mixed-effects models, using longer time-series data to enhance robustness and capture dynamic firm-level heterogeneity. Future research could utilise a dependent variable with a measure that is derived from an exclusively sustainability reporting standard such as the GRI Guidelines and the TCFD. This could allow for a more comprehensive assessment of SRQ. Future studies should consider extending the analysis to smaller firms to provide a more comprehensive understanding of sustainability reporting across different organisational contexts. Furthermore, future studies can conduct a qualitative analysis to understand the underlying factors associated with the relationship between diversity variables such as ethnicity and age and the SRQ of a firm. Beyond quantitative analysis, deeper insights can be derived from case studies and interviews with members of diverse teams.

Author Contributions

Conceptualization, T.N.; methodology, O.T.; software, O.T.; validation, T.N., K.M., and O.T.; formal analysis, O.T.; investigation, K.M.; resources, K.M.; data curation, K.M.; writing—original draft preparation, T.N.; writing—review and editing, K.M.; visualization, T.N.; project administration, T.N. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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